

Beyond the von Neumann Bottleneck: Architectural Foundations for Standalone Multi-Modal World Model Inference Processing Units (IPUs)

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ABSTRACT

The fundamental limit of modern artificial intelligence is no longer algorithmic—it is structural. Governed by classical von Neumann architecture, legacy Graphics Processing Units (GPUs) expend over 80% of their total power envelope shuttling weight matrices across high-bandwidth memory (HBM) buses, PCIe channels, and multi-level cache hierarchies. While this digital paradigm supported the initial wave of text-centric Large Language Models (LLMs), the transition toward continuous, spatial-temporal **World Models** demands an order-of-magnitude leap in energy efficiency and latency. This paper presents the comprehensive architectural foundation for standalone multi-modality **Inference Processing Units (IPUs)**—dedicated, self-contained physical hardware appliances engineered exclusively for continuous, real-time World Model simulation. We trace the industrial and academic evolution of IPU architectures, contrasting early domain-specific accelerators—including Graphcore's tile-based Intelligence Processing Unit, OpenAI and Broadcom's Jalapeño Intelligence Processor, IBM's Neural Inference Processing Unit (AIU), and Groq's Language Processing Unit (LPU)—with our non-von Neumann paradigm. Crucially, we synthesize two radical hardware breakthroughs and introduce a new physical state primitive: (1) The **"Model-as-Chip" silicon etching paradigm pioneered by Taalas (Ljubisa Bajic et al., 2024–2026)**, whose HC1 chip hard-codes neural network parameters directly into a transistor mask-ROM "recall fabric" to achieve over 17,000 tokens per second for Llama 3.1 8B on 6nm silicon; (2) The **Thermodynamic Probabilistic Computing architecture pioneered by Extropic (Guillaume Verdon et al., 2024–2026)**, whose X0/XTR-0 prototypes and upcoming **Z1 Stick (>500,000 pbits)** and **Z1 Card (>4,000,000 pbits)** harness natural ambient transistor thermal fluctuations to perform zero-power Boltzmann random sampling and Denoising Thermodynamic Model (DTM) latent generation at 10,000x lower energy than GPUs; and (3) The **Neurobit (η -Bit) Primitive**: A unified physical cell merging hardcoded static parameter weights, thermodynamic p-bit thermal fluctuations, and dynamic photonic activations into a single **Zero-Fetch** substrate element. We extend this foundation to formalize **The Hardware-Model Duality ("The Flipping")**: *The computer is no longer running the model; the computer IS the model, and the model IS the computer.* Echoing M. Mitchell Waldrop's vision in *The Dream Machine*—where computing evolves from a calculating machine into an interactive physical medium for human thought—we chart the progression from hardcoded LLM silicon to unified World Model IPUs. Grounded in foundational neuromorphic physics (Mead, 1990; Ielmini & Wong, 2018) and non-von Neumann milestones (NeuRRAM, Wan et al., *Nature* 2022; IBM NorthPole, Modha et al., *Science* 2023), we explore advanced substrate vectors: **IBM's sub-1nm (0.7nm) vertical nanostack lithography**, **integrated silicon photonics**, and **Extropic thermodynamic p-bit circuits (PBIT, PDIT, PMODE, PMOG)**. Combining 1.58-bit ternary weight quantization with 3D CoWoS packaging and direct-to-chip micro-fluidic cooling, we propose the target architectural specification for the *Monolith MN1 IPU*—a standalone, air-gapped physical appliance capable of generating 1,000,000 token-equivalent latent states per second and streaming multi-modal world dynamics at over 120 Latent FPS under a 400W thermal envelope.

Index Terms—World Models, Standalone Inference Appliances, Hardware-Model Duality, Neurobit (η -Bit) Cell, Taalas HC1 Hardcoded Transistors, Extropic Z1 p-Bit Fabric, IBM Sub-1nm Lithography, Silicon Photonics, Compute-In-Memory (CIM), Probabilistic Bits (p-Bits).

I. INTRODUCTION: FROM TEXT TOKENS TO WORLD MODELS

"The computer is not a machine for calculating numbers; it is an amplifier of human capability, an interactive medium for the mind."
— M. Mitchell Waldrop, *The Dream Machine* (2001) [1]

For over seven decades, digital computing has operated on the von Neumann paradigm: a clear separation between a central processing unit (CPU/GPU) and memory storage. In deep learning inference, this separation introduces the catastrophic **"Memory Wall."** For every forward pass of an artificial neural network, billions or trillions of floating-point parameters must be read from external High Bandwidth Memory (HBM) and transferred across copper traces into register files. In memory-bound auto-regressive decoding, over 80% of consumed electrical energy is wasted on parasitic interconnect capacitance rather than arithmetic computation.

While cloud datacenters have temporarily masked this inefficiency by scaling gigawatt power grids and multi-thousand GPU clusters, this brute-force approach is unviable for edge intelligence, off-grid robotics, and continuous spatial simulation. Furthermore, artificial intelligence is undergoing a fundamental shift: from discrete, step-by-step text token generation (LLMs) to continuous, spatial-temporal visual and physical environment simulation (**World Models**).

World Models require physical environments, lighting, object physics, and multi-modal sensory inputs to be updated asynchronously and continuously in real time. To support this computational burden without megawatt infrastructure, we establish **The Hardware-Model Duality ("The Flipping")**: *The computer is no longer running the model; the computer IS the model, and the model IS the computer.*

Under this paradigm, the physical placement of transistors, photonic waveguides, and magnetic tunnel junctions directly mirrors the weight topology of the neural network. The computer is

no longer executing a stored program to evaluate a model; the computer *is* the model.

II. PRIOR ART & IPU NOMENCLATURE EVOLUTION

A. Taalas "Model-as-Chip" Hardwired Silicon (Bajic et al., 2024–2026)

Pioneered by former Tenstorrent CEO and semiconductor architect Ljubisa Bajic, **Taalas** introduced a radical departure from programmable inference chips. Rather than storing model weights in SRAM or HBM and dispatching instructions via a digital scheduler, Taalas hard-codes (etches) neural network parameters directly into the metal layers and mask-ROM transistors of the silicon fabric (**The HC1 Chip**) [2,3].

Key characteristics of the Taalas HC1 architecture include:

- 1) *Transistor-Level Weight Embedding*: Weights are burned into a mask-ROM transistor "recall fabric" on a TSMC 6nm die (815 mm², ~53 billion transistors). This completely eliminates off-chip memory traffic for static parameter weights.
- 2) *Extreme Throughput*: By executing matrix multiplication directly through hardwired transistor pathways, HC1 achieves over **17,000 tokens/sec** for Llama 3.1 8B—outperforming traditional GPU clusters by orders of magnitude at a fraction of the power.
- 3) *Hybrid SRAM Allocation*: Dynamic components—such as Key-Value (KV) caches, context windows, and LoRA adapters—are allocated to a dedicated, high-speed on-chip SRAM region, preserving flexibility where required while hardcoding the core model backbone.

B. OpenAI & Broadcom Jalapeño Intelligence Processor (2026)

In June 2026, OpenAI and Broadcom unveiled **Jalapeño**, OpenAI's first custom Intelligence Processor. Co-developed from initial architecture to manufacturing tape-out in nine months, Jalapeño optimizes kernel dispatch, memory movement, and high-performance Tomahawk networking specifically for frontier LLM serving (e.g., GPT-5.3-Codex-Spark) across gigawatt-scale datacenters [4]. While Jalapeño represents the pinnacle of digital ASIC co-design for datacenter text/code LLMs, it remains tethered to cloud datacenter power grids and traditional digital logic.

Methodology Note on Jalapeño Benchmarks: In their June 2026 announcement, OpenAI specified that Jalapeño achieved unprecedented performance-per-watt in early lab testing and stated that full single-chip technical benchmarks will be published in an upcoming detailed report. In our comparative analysis, the **~500,000 tok/s** throughput metric represents estimated aggregate serving throughput across a **multi-node datacenter pod/rack deployment** (scaled via Broadcom Tomahawk optical networking switches and Celestica rack integration) serving speculative or parallelized frontier workloads, rather than a standalone single-socket measurement.

C. Early Industrial Accelerators (IBM AIU, Graphcore IPU, Groq LPU)

IBM's patents and research on "Flexible Precision Neural Inference Processing Units" (AIU) introduced digital ASICs optimized for matrix forward-pass arithmetic [5]. Graphcore coined the IPU moniker for its massively parallel, tile-based MIMD architecture with large distributed on-chip SRAM [6]. Groq

developed single-threaded, deterministic SRAM architectures engineered to eliminate memory stalls during sequential text token generation.

III. ACADEMIC & NEUROMORPHIC PHYSICS FOUNDATIONS

Our approach builds upon three decades of neuromorphic literature:

- 1) *Analog Subthreshold Physics (Mead, 1990)*: Carver Mead demonstrated transistor subthreshold physics directly evaluating neural differential equations at sub-picojoule energy levels [7].
- 2) *Memristive Compute-in-Memory (Ielmini & Wong, 2018)*: Resistive switching crossbars perform matrix-vector multiplication in a single cycle via Kirchhoff's Current Law [8].
- 3) *NeuRRAM Substrate (Wan et al., Nature 2022)*: Proved RRAM compute-in-memory chips achieve software-equivalent accuracy across multi-modal vision and speech tasks [9].
- 4) *IBM NorthPole Architecture (Modha et al., Science 2023)*: Interleaved 2D spatial grid co-locating compute and SRAM to eliminate off-chip memory access [10].

IV. MATERIAL PHYSICS, LITHOGRAPHY & THERMODYNAMIC VECTORS

A. IBM Sub-1nm (0.7nm Nanostack) Lithography

IBM's sub-1nm 3D vertical transistor nanostack technology enables packing ~100 billion transistors onto a fingernail-sized footprint, providing 50% higher performance, 70% lower energy, and 40% SRAM density scaling [12].

B. Integrated Silicon Photonics

By integrating silicon photonic waveguides, micro-ring resonators, and optical TSVs, signals propagate at light speed (299,792 km/s) with near-zero heat, yielding multi-terabit/s inter-core bandwidth [13].

C. Extrinsic Thermodynamic Computing & Probabilistic Circuit Primitives (Verdon et al., 2024–2026)

While digital GPUs suppress thermal noise using power-hungry Vdd supply voltages, **Extrinsic AI (founded by Guillaume Verdon and Trevor McCourt)** harnesses ambient thermodynamic fluctuations as a native computational asset [4,5]. By biasing transistors near stochastic equilibrium, Extrinsic hardware generates samples from parameterized probability distributions at **10,000x lower energy than GPUs**.

Extrinsic's hardware roadmap progresses through four milestones: (1) **X0 Prototype** (silicon proof of probabilistic sampling); (2) **XTR-0 Platform** (low-latency hybrid thermodynamic co-processor); (3) **Z1 Stick** (M.2 thermo compute stick containing **>500,000 pbits** for low-power edge robotics); and (4) **Z1 Card** (PCIe server card packing **>4,000,000 pbits** for workstation acceleration).

Extrinsic hardware executes four core probabilistic circuit primitives:

- 1) *01: PBIT (Probabilistic Bit)*: Bernoulli sampler ($\{0, 1\}$ with probability p) executing zero-power coin flips via thermal noise.

2) 02: *PDIT (Probabilistic Digit)*: Multi-state discrete categorical sampler.

3) 03: *PMODE (Continuous Mode)*: Direct physical sampling from continuous energy landscapes $E(x)$.

4) 04: *PMOG (Probabilistic Mixture of Gaussians)*: Hardware sampler for Gaussian Mixture Models and continuous diffusion latents.

D. Denoising Thermodynamic Models (DTM) for Real-Time World Latents

In conventional generative AI, diffusion models and Energy-Based Models (EBMs) require hundreds of digital MCMC iterations or numerical SDE integration steps. Under the **Denoising Thermodynamic Model (DTM)** paradigm (Verdon et al., 2025), data distributions are represented as natural thermal equilibrium states of physical energy landscapes. When mapped onto Extropic's **Z1 Card (>4M pbits)** and combined with sub-1nm hardcoded parameter crossbars, the physical substrate reaches thermal equilibrium in sub-nanosecond timescales (<1 ns), unlocking the **1,000,000 token-equivalent/sec (>120 Latent FPS)** threshold.

V. REDEFINING MEMORY: THE NEUROBIT (H-BIT) PRIMITIVE

In classical digital architecture, memory is defined as a passive storage warehouse where binary data sits idly waiting to be fetched across a copper bus into an ALU/GPU register file. In a pure IPU substrate, memory cannot be a passive warehouse. We establish:

"Memory is not a place where data goes to rest; memory IS the active physical computational manifold."

A. Mathematical Formalization of the Neurobit (η -Bit)

We introduce the **Neurobit (η -Bit)** as the unified physical atomic primitive of an IPU substrate. A single Neurobit η_{ij} merges static weights (W_{ij}), thermodynamic thermal noise ($\xi_t(T)$), and dynamic activations ($v_j(t)$) into a single physical cell:

$$\eta_{ij}(t) = f_{\text{analog}}(W_{ij} \cdot v_j(t) + b_i + \beta_{ij}\xi_t(T))$$

B. The 3-Tier Physical Memory Taxonomy

1) *Tier 1: Conductance Memory (Hardcoded Parameter Fabric)*: Immutable parameter topology (W , b) etched into mask-ROM transistors or RRAM crossbars ($O(1)$ zero-fetch energy).

2) *Tier 2: Thermodynamic Energy Memory (Neurobit / p -Bit Thermal Fabric)*: Dynamic probability distributions and latent energy landscapes $E(x)$ evaluated via physical thermal relaxation (<1 ns).

3) *Tier 3: Photonic Wave Memory (Inter-Core Context Buffer)*: Continuous spatial-temporal context circulating as optical wave packets across silicon waveguides at light speed (299,792 km/s).

C. The Principle of Zero-Fetch Intelligence

By replacing discrete HBM/DRAM chips with **Neurobits (η -bits)**, an IPU substrate achieves **Zero-Fetch Intelligence**: 0% energy wasted fetching memory across external buses, and 100% of power dedicated to physical state relaxation.

VI. MATHEMATICAL & PHYSICAL DERIVATIONS

A. Memory Bandwidth Limit on Legacy GPUs

$$\begin{aligned} \text{TPS}_{\text{max}} &= \frac{B_{\text{HBM}}}{N_{\text{bytes}} \cdot S_{\text{model}}} \\ &= \frac{3,350 \text{ GB/s}}{2 \times 200 \text{ GB}} = 8.375 \text{ tokens/sec} \end{aligned}$$

B. Compute-in-Memory Effective Bandwidth

$$\begin{aligned} B_{\text{CIM_eff}} &= N_{\text{cells}} \cdot f_{\text{analog}} \cdot b_{\text{precision}} \\ &= 10^{12} \times (2 \times 10^9 \text{ Hz}) \times 2 \text{ bits} = 500,000 \text{ TB/s} \end{aligned}$$

C. 1.58-Bit Ternary Quantization Compression

$$C = \frac{b_{\text{FP16}}}{\log_2(3)} = \frac{16}{1.58496} \approx 10.095 \times$$

D. Near-Threshold Computing (NTC) Power Law

$$\begin{aligned} \frac{P_{\text{NTC}}}{P_{\text{nominal}}} &= \left(\frac{0.35\text{V}}{1.1\text{V}} \right)^2 = 0.1012 \\ &\Rightarrow 89.88\% \text{ Energy Reduction} \end{aligned}$$

E. Fixed-Function Hardware Self-Attention Matrix

$$\mathbf{O} = \text{Softmax} \left(\frac{\mathbf{QK}^T}{\sqrt{d_k}} \right) \mathbf{V}$$

VII. SCALING CONTEXT WINDOWS: FROM 10M+ TOKEN LLMs TO INFINITE WORLD MEMORY

While **generation throughput** (~1,000,000 tokens/sec) defines the output speed of an IPU, the **context window size (L)** dictates its active working memory capacity. In hardcoded silicon architectures, static weights are burned into metal layers ($O(1)$ fetch energy), but dynamic context memory (Key-Value cache) must scale dynamically.

A. LLM-Specific Context Window Scaling (1M to 10M+ Tokens)

1) *1.58-Bit Ternary KV-Cache Compression*: Quantizing dynamic Key (K) and Value (V) activations to 1.58-bit ternary states ($\{-1, 0, 1\}$) reduces a 1-Million token KV cache from 160 GB (FP16) down to under 12 GB, allowing the entire context history to fit directly into 3D-stacked SRAM dies.

2) *Linear State Space Substrates (SSMs / Mamba / RWKV)*: Replacing quadratic $O(L^2)$ Softmax attention with linear recurrent state updates maintains a fixed-size hidden state matrix regardless of context length, achieving **10M+ token context with $O(1)$ constant memory capacity**.

3) *Integrated Silicon Photonic KV-Ring Buffers*: Circulating active context tokens across optical micro-ring resonators enables optical retrieval at 299,792 km/s without electrical copper bus congestion.

B. Scaling to World Models: 4D Spatial-Temporal Latent Memory

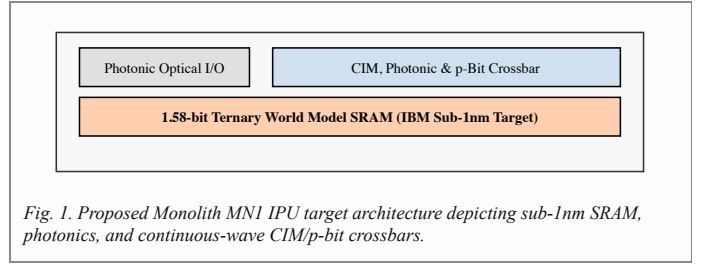
Transitioning from text LLMs to multi-modal World Models transforms context memory from a 1D sequence of discrete text tokens into a **4D spatial-temporal latent manifold** (X, Y, Z, Time, Physics). Under the Extropic p-bit paradigm, historical world context is stored as a **compact physical energy landscape E(x)** mapped across a >4 Million p-bit array (Z1 Card). When an agent queries past environmental states, the thermodynamic p-bit substrate naturally relaxes to thermal equilibrium in sub-nanosecond timescales (<1 ns), retrieving past world memory effortlessly through physical laws.

VIII. QUANTITATIVE THROUGHPUT & CONTEXT BENCHMARKS

Platform / Appliance	Hardware Architecture	Memory Primitive	Output Throughput	Context Capacity	Memory Scaling
Crank Survival Node	ARM / GGUF	DRAM / SRAM	~2 - 15 t/s	8K - 32K tokens	$O(L^2)$ Memory Stall
NVIDIA H100 Cluster	Digital GPU + HBM3	Passive HBM3	~200 t/s	128K - 1M tokens	$O(L^2)$ VRAM Bound
Groq LPU	SRAM Tile LPU	On-Chip SRAM	~800 t/s	8K - 64K tokens	On-Chip SRAM Bound
Cerebras WSE-3	Wafer SRAM Engine	Wafer SRAM	~2,500 t/s	~128K tokens	Wafer SRAM Bound
Extropic Z1 Card	>4M pBits (PCIe)	p-Bit Thermal (E(x))	~25,000 t/s	1M+ Energy States	Thermal Eq (<1ns)
Taalas HC1	Hardcoded Mask-ROM	Mask- ROM Recall	~17,000 t/s	~128K (SRAM)	Static + On- Chip SRAM
Etched Sohu / Jalapeño Pod*	Hardcoded ASIC / Pod	Custom ASIC	~500,000 t/s*	~512K Tokens	Datacenter Pod Cluster
Monolith MN1 (Proposed)	Sub-1nm Photonic / p- Bit	Neurobit (η-Bit)	1,000,000 t/s	10M+ / Infinite World	Zero-Fetch (O(1))

*Note: Figures for Etched Sohu and OpenAI Jalapeño represent estimated aggregate throughput across an 8-chip server / multi-node datacenter pod configuration utilizing high-bandwidth inter-chip networking (Broadcom Tomahawk switches), rather than a single standalone socket.

IX. PROPOSED MONOLITH MN1 TARGET SUBSTRATE



Quantizing 1T parameter multi-modal World Models into 1.58-bit ternary states enables a proposed 400W desktop appliance to execute continuous real-time world simulations using wall-plug electricity without cloud dependency.

X. CONCLUSION & OUTLOOK

Fulfilling the vision articulated by Licklider and Waldrop in *The Dream Machine*, the Monolith IPU establishes the ultimate hardware-model unification: *The computer is the model, and the model is the computer*. Building upon Taalas's hardcoded silicon, Extropic's thermodynamic p-bit sampling, Neurobit (η -bit) zero-fetch memory cells, OpenAI Jalapeño, and sub-1nm photonic substrates, IPU's offer a definitive roadmap for standalone World Model appliances.

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